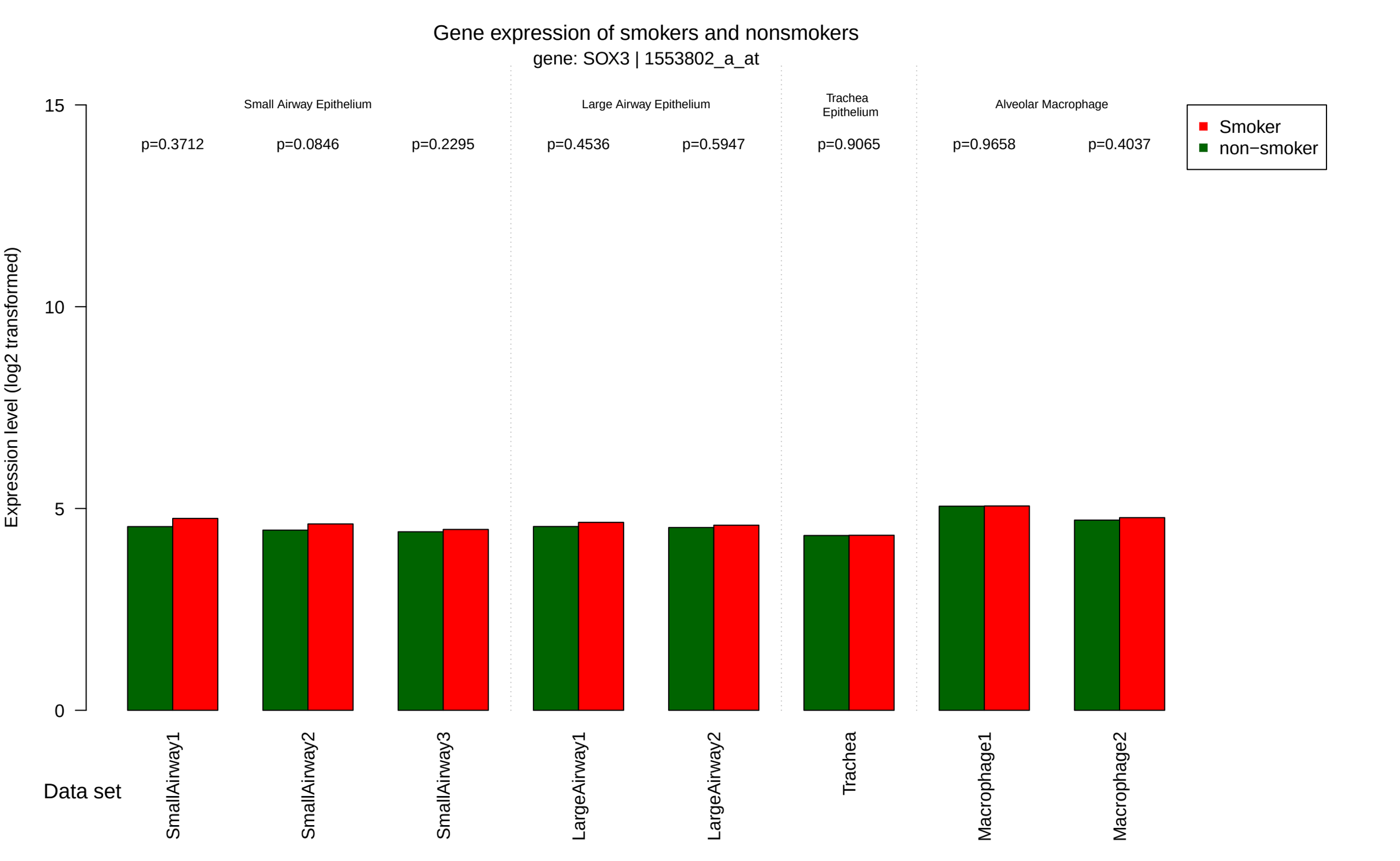
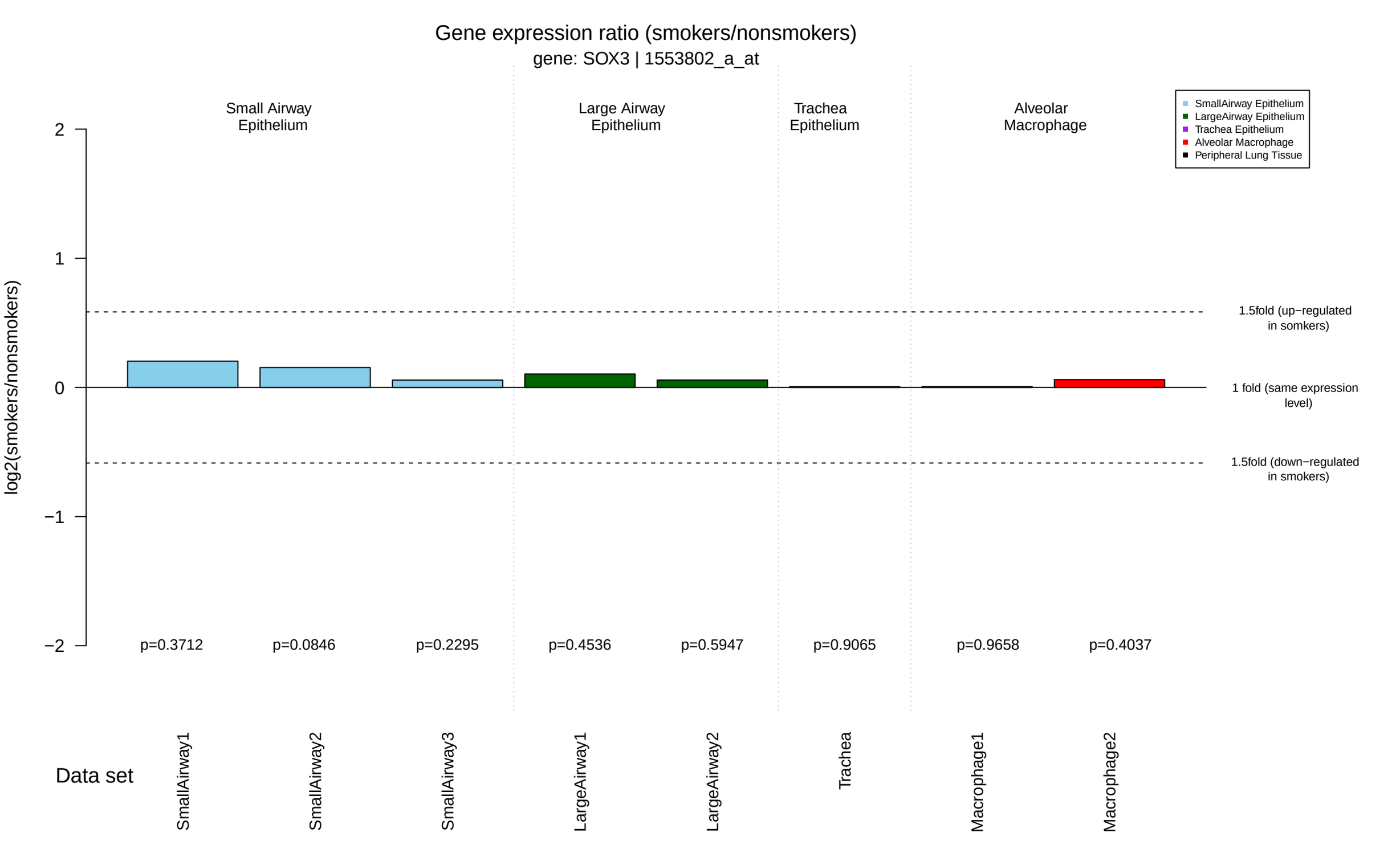
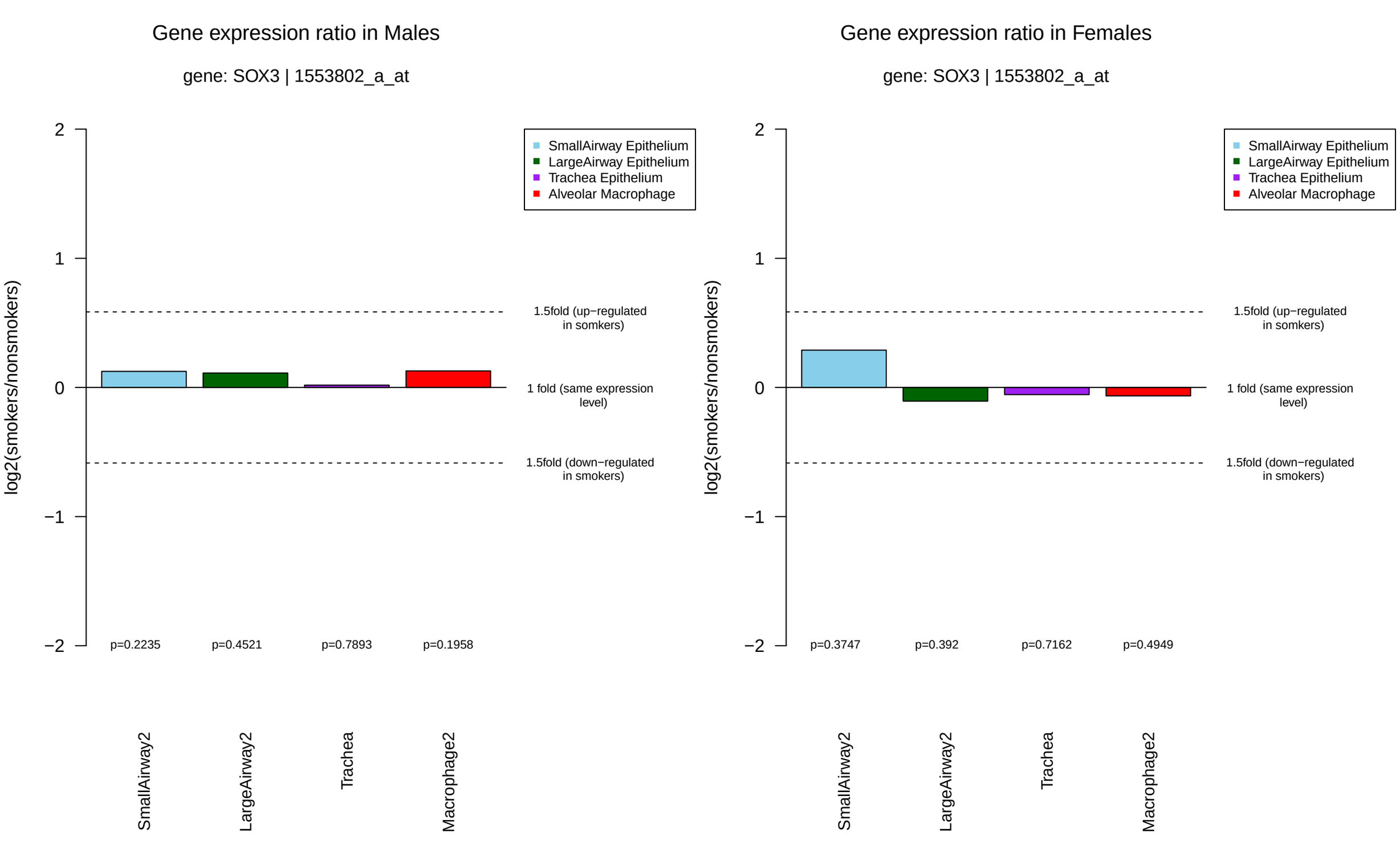
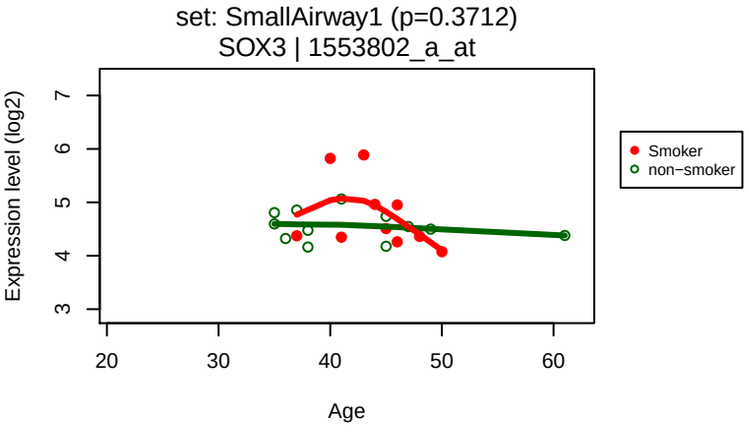
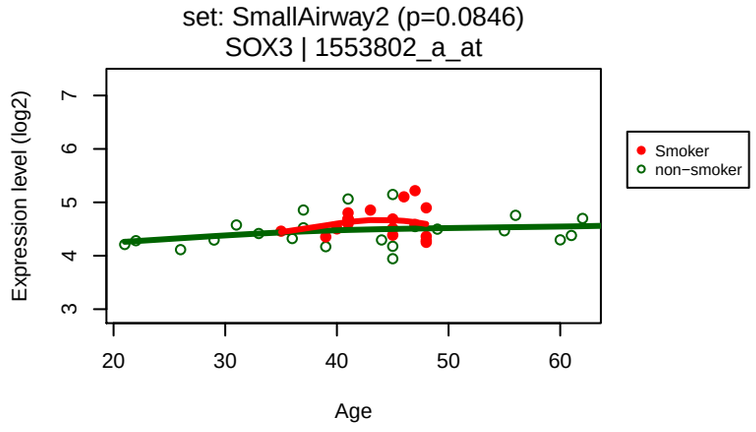
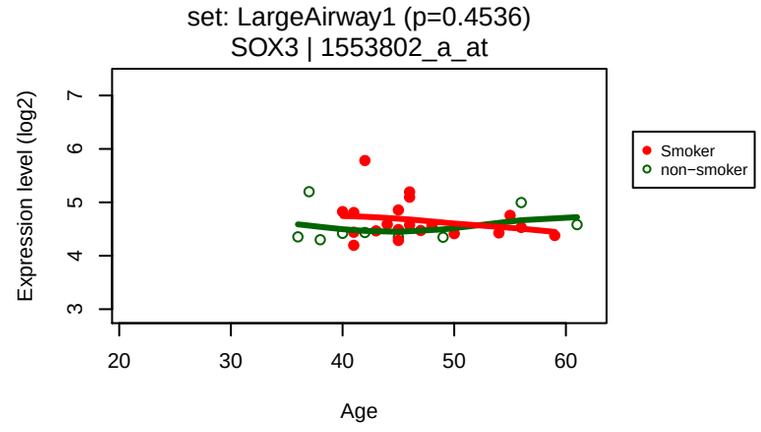

Gene expression of smokers and nonsmokers
gene: SOX3 | 1553802_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3712
p=0.0846
p=0.2295
p=0.4536
p=0.5947
p=0.9065
p=0.9658
p=0.4037
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

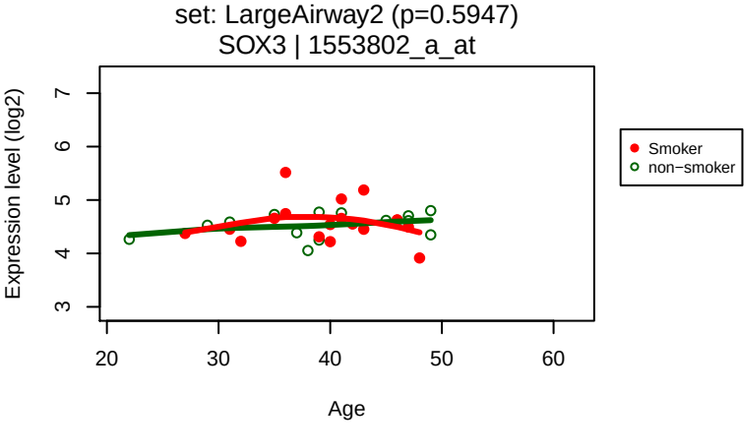

Gene expression ratio (smokers/nonsmokers)
gene: SOX3 | 1553802_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3712
p=0.0846
p=0.2295
p=0.4536
p=0.5947
p=0.9065
p=0.9658
p=0.4037
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

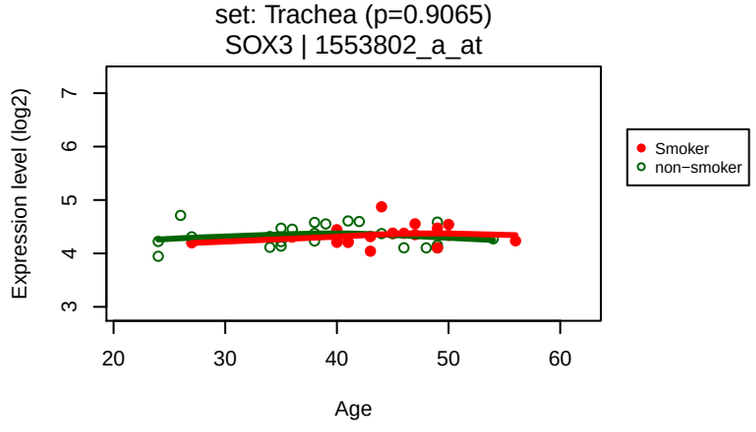

Gene expression ratio in Males
gene: SOX3 | 1553802_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2235
p=0.4521
p=0.7893
p=0.1958
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SOX3 | 1553802_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3747
p=0.392
p=0.7162
p=0.4949
SmallAirway2
LargeAirway2
Trachea
Macrophage2

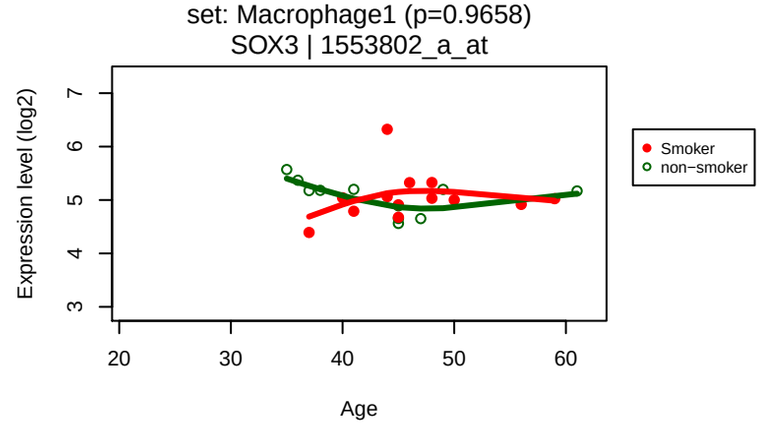

set: SmallAirway1 (p=0.3712)
SOX3 | 1553802_a_at
Expression level (log2)
Age
Smoker
non−smoker

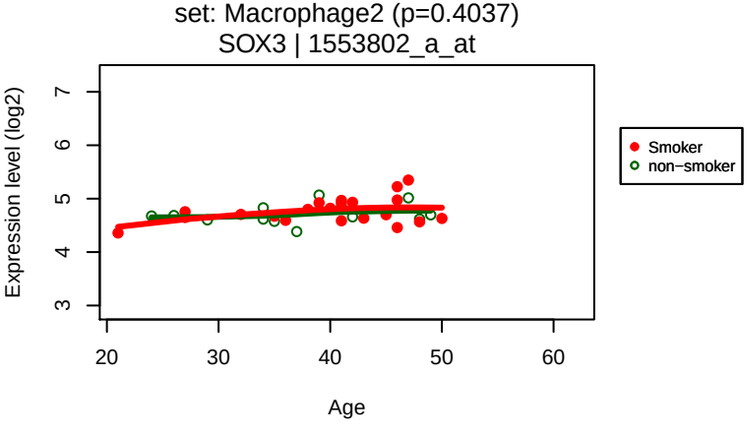

set: SmallAirway2 (p=0.0846)
SOX3 | 1553802_a_at
Expression level (log2)
Age
Smoker
non−smoker

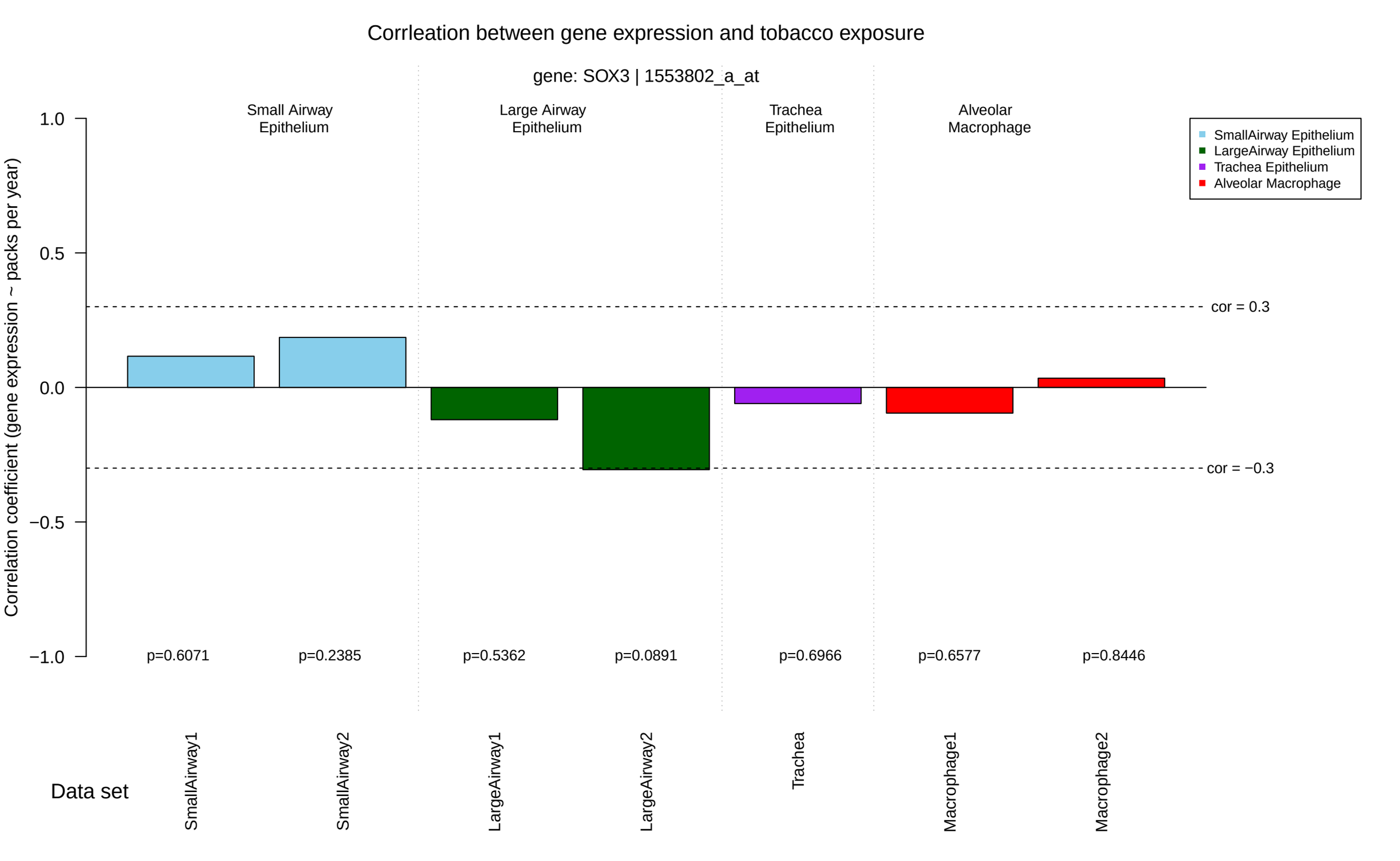

set: LargeAirway1 (p=0.4536)
SOX3 | 1553802_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5947)
SOX3 | 1553802_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9065)
SOX3 | 1553802_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9658)
SOX3 | 1553802_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4037)
SOX3 | 1553802_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SOX3 | 1553802_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6071
p=0.2385
p=0.5362
p=0.0891
p=0.6966
p=0.6577
p=0.8446
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2